

DATA ANALYSIS REPORT

Cairo Housing Market Analysis

Factors Influencing Residential Property Prices in Cairo, Egypt

Course	C-DE211: Data Analysis
Project	Project 3 — Final Data Analysis Project
Dataset	Egypt Cairo Housing Prices (Kaggle — 11,418 listings)
Clean Dataset	9,209 rows after preprocessing
Team	Team GB : Eyad Mohamed Fakeih Abdurrahman Ehab Kareem Ali Ali Ahmed

1. Introduction

The Egyptian real estate market, particularly in Cairo, has experienced remarkable growth and complexity over the past decade. Driven by rapid urbanization, population growth, and the development of new urban communities such as New Cairo and the New Administrative Capital, residential property prices in Cairo have become increasingly dynamic and multifaceted. Understanding the key factors that drive property prices is of critical importance to homebuyers, real estate investors, urban planners, and policymakers alike.

This report presents a comprehensive data analysis of the Cairo residential property market, conducted using a real-world dataset of over 11,000 property listings scraped from an active Egyptian real estate marketplace. The analysis covers the full pipeline of a data science project: data collection, preprocessing, exploratory data analysis, and formal statistical hypothesis testing.

The central research question guiding this study is:

"What are the key factors that significantly influence residential property prices in Cairo, Egypt?"

To answer this question, three specific hypotheses were formulated and tested using appropriate statistical methods, providing quantitative evidence about the relationships between property characteristics and market price.

2. Dataset Description

The dataset used in this study is the Egypt Cairo Housing Prices dataset, publicly available on Kaggle (kaggle.com/datasets/iyadelwy/egypt-housing-prices). It was collected by scraping active residential property listings from a major Egyptian real estate marketplace, providing a realistic snapshot of Cairo's current housing market.

2.1 Dataset Overview

Attribute	Details
Original Records	11,418 property listings
Features	7 columns
Geographic Scope	Cairo, Egypt (100% Cairo-based)
Missing Values	None — dataset is complete
After Cleaning	9,209 rows (removed duplicates, invalid prices, outliers)

2.2 Feature Descriptions

Column	Type	Description
<code>type</code>	Categorical	Property type: Apartment, Villa, Duplex, Townhouse, Penthouse, Twin House, iVilla, Hotel Apartment, Chalet
<code>title</code>	Text	Listing title as posted by the seller on the marketplace
<code>location</code>	Categorical	Full address including compound, district, and city area within Cairo
<code>bedroom</code>	Ordinal	Number of bedrooms (0=Studio, 1–8)
<code>bathroom</code>	Integer	Number of bathrooms in the property
<code>size_sqm</code>	Numeric	Property area in square meters — cleaned to 1–487 m ²
<code>price</code>	Numeric	Listing price in Egyptian Pounds (EGP) — target variable

3. Methodology

3.1 Data Preprocessing

The raw dataset required several cleaning steps before analysis could begin:

- **Duplicate Removal:** 308 exact duplicate rows were identified and removed.
- **Price Conversion:** The price column was stored as text (e.g., "6,850,000"). Commas were stripped and the column was converted to integer. 402 listings with price = "Ask" (no real price posted) were dropped as they provide no analytical value.
- **Size Conversion:** The size_sqm column was similarly stored as text and converted to numeric integer format.
- **Bedroom Cleaning:** The bedroom column contained "Studio" values and one malformed entry ({0}). Studio was recoded as 0 and malformed entries were removed.
- **Outlier Removal:** The IQR method ($1.5 \times \text{IQR}$) was applied to both price and size_sqm, removing 1,499 rows containing extreme outliers. Final valid price range: 1M–16.25M EGP. Final valid size range: 1–487 m².

After preprocessing, the final clean dataset contained 9,209 rows with zero missing values and correctly typed columns.

3.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the distributions, relationships, and patterns within the data prior to formal hypothesis testing. Visualizations included distribution plots, boxplots, bar charts, line charts, and a correlation heatmap.

3.3 Hypothesis Testing

Three hypotheses were formulated and tested at a significance level of $\alpha = 0.05$ using the following statistical tests:

- H1 — Independent Samples T-Test: Used to compare the mean price of two independent groups (Villas vs. Apartments). This test is appropriate when comparing a numeric variable across two categorical groups.
- H2 — Pearson Correlation: Used to measure the linear relationship between two numeric variables (size_sqm and price). The coefficient r ranges from -1 to +1.
- H3 — One-Way ANOVA: Used to compare mean prices across three or more groups (bedroom counts 0–6). ANOVA tests whether at least one group mean differs significantly from the others.

All tests were conducted using the SciPy library in Python. Results were interpreted with supporting visualizations for each hypothesis.

4. Exploratory Data Analysis

4.1 Property Type Distribution

The dataset contains 10 distinct property types. As shown in Figure 1, Apartments dominate the Cairo market with 5,493 listings (59.6% of all properties), followed by Villas (13.5%) and Townhouses (8.6%). This distribution reflects Cairo's urban density where most residents live in multi-unit apartment buildings. The remaining property types — Duplex, Twin House, Penthouse, iVilla, Hotel Apartment, Chalet, and Compound — collectively account for approximately 27% of listings.

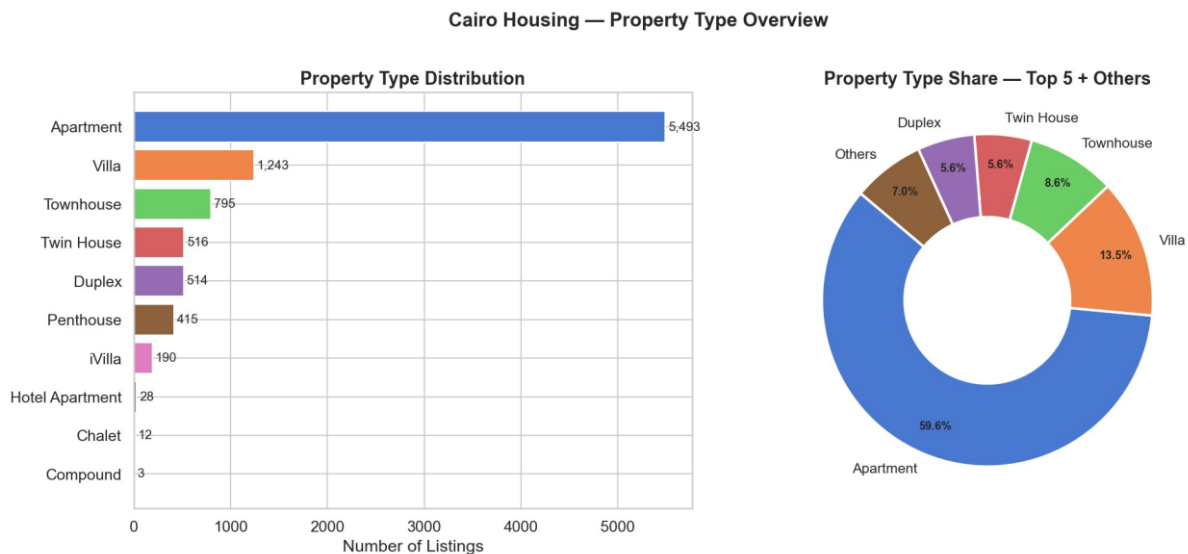


Figure 1: Property type distribution — bar chart and donut chart showing market share

4.2 Price Distribution

The price distribution (Figure 2) exhibits a strong right skew, typical of real estate markets. The majority of listings cluster between 1M and 5M EGP, with a long tail extending to 16M EGP. The median price (3.5M EGP) lies significantly below the mean, confirming the right-skewed nature of the data. The boxplot reveals that the interquartile range spans approximately 2.5M to 6M EGP, representing the core mid-market segment.

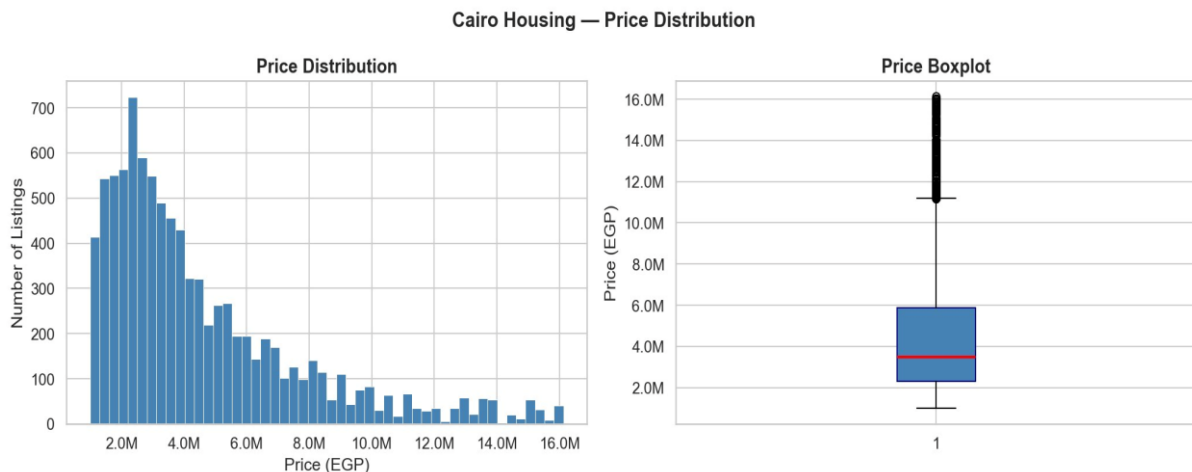


Figure 2: Price distribution — histogram (left) and boxplot (right)

4.3 Property Size Distribution

The size distribution (Figure 3) shows that most Cairo properties fall between 130 and 280 m², with a peak around 160–170 m². The median size is approximately 210 m². The distribution is slightly right-skewed, with some larger properties extending to 487 m² after outlier removal. This range is consistent with typical Cairo residential properties, from compact apartments to larger villas.

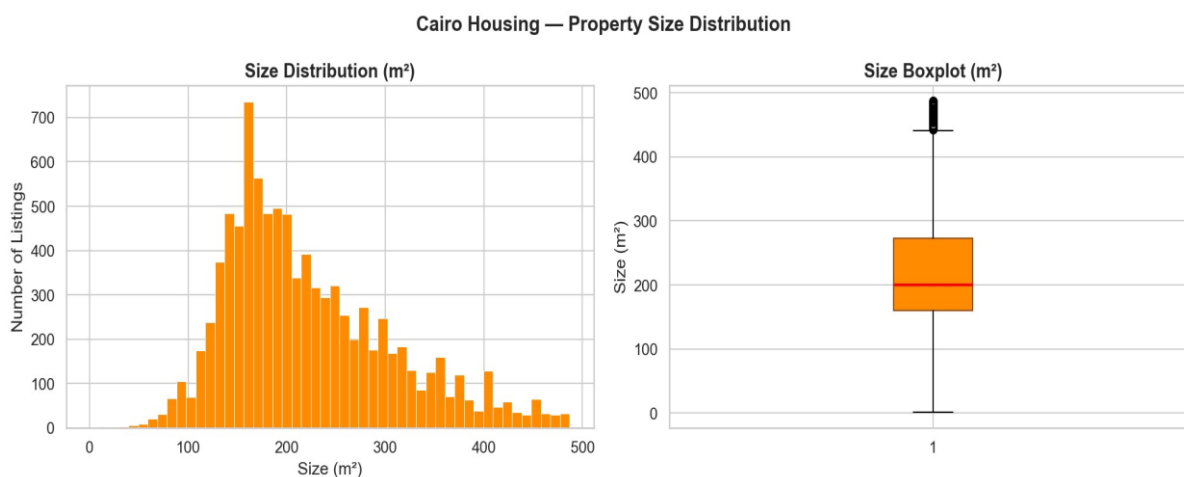


Figure 3: Property size distribution (m²) — histogram (left) and boxplot (right)

4.4 Average Price by Property Type

Figure 4 reveals a clear price hierarchy across property types. Twin Houses (9.13M EGP) and Villas (8.14M EGP) command the highest average prices, nearly three times the average Apartment price (3.13M EGP). This large gap directly motivates Hypothesis 1. Townhouses (6.14M EGP) occupy a middle ground, while Apartments, Hotel Apartments, and Chalets represent the more affordable end of the market.

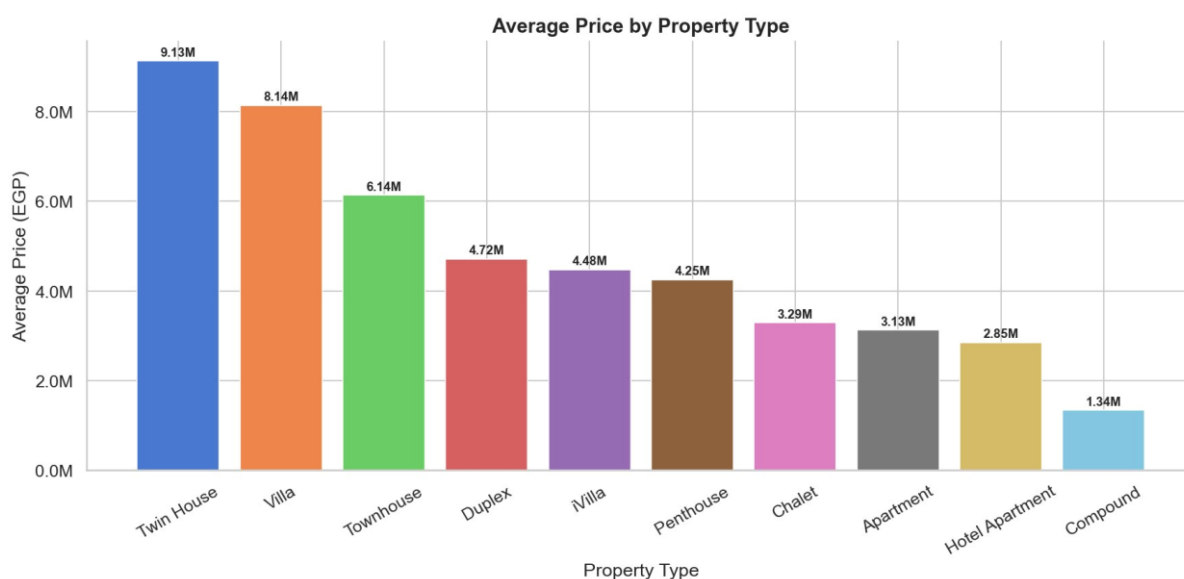


Figure 4: Average listing price by property type (EGP)

4.5 Price Spread per Property Type

The boxplot in Figure 5 shows not just averages but the full price spread within each property type. Twin Houses and Villas display wide interquartile ranges, indicating high variability — some are relatively affordable while others are luxury properties. Apartments, by contrast, show a tight, compact distribution reflecting their consistent, affordable positioning in the market. Compound listings show almost no variation due to having only 3 listings.

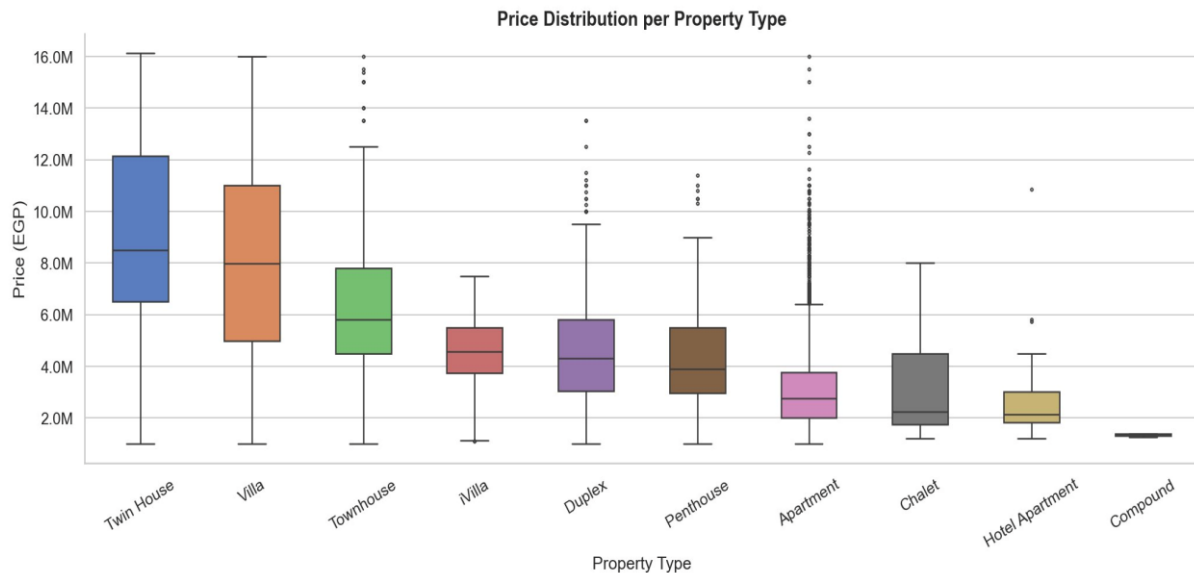


Figure 5: Price distribution per property type — sorted by median price

4.6 Bedroom Count Distribution

Figure 6 shows that 3-bedroom properties are overwhelmingly the most common (5,210 listings, 56.7%), reflecting the typical Cairo family home. 4-bedroom properties are second (1,891 listings), followed by 2-bedroom units (1,391). Studios and properties with 7 or more bedrooms are extremely rare, suggesting these represent niche market segments. The concentration in the 2–4 bedroom range reflects Cairo's housing preferences for family-sized properties.

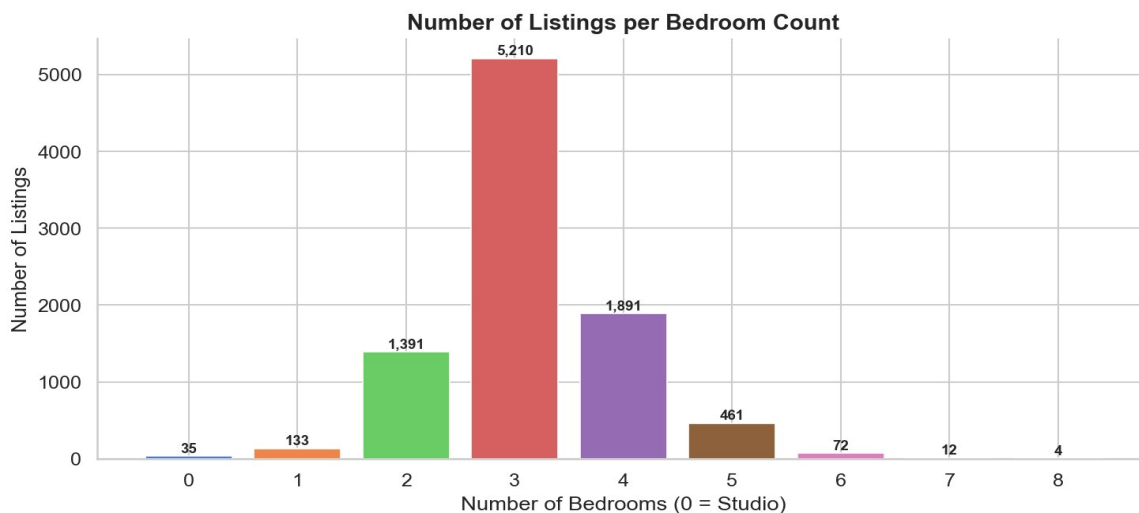


Figure 6: Number of listings per bedroom count (0 = Studio)

4.7 Average Price by Bedroom Count

A clear upward trend is visible in Figure 7, where mean property prices increase consistently with bedroom count. Studios average 1.97M EGP while 5-bedroom properties average 7.53M EGP — a nearly 4x increase. The slight drop at 6 and 8 bedrooms is attributable to the very small sample sizes in those groups (72 and 4 listings respectively), which makes those averages less statistically reliable. This trend strongly motivates Hypothesis 3.



Figure 7: Average property price by number of bedrooms

4.8 Correlation Heatmap

The correlation heatmap (Figure 8) reveals the pairwise relationships between all numeric features. The strongest correlation with price is size_sqm ($r = 0.66$), confirming that property size is the dominant numeric predictor of price. Bathroom count also shows a moderate correlation with price ($r = 0.53$), while bedroom count shows a moderate positive relationship ($r = 0.44$). Notably, bedroom and bathroom counts are highly correlated with each other ($r = 0.71$) and with size ($r = 0.69$ and 0.68 respectively), which is expected as larger properties naturally have more rooms.

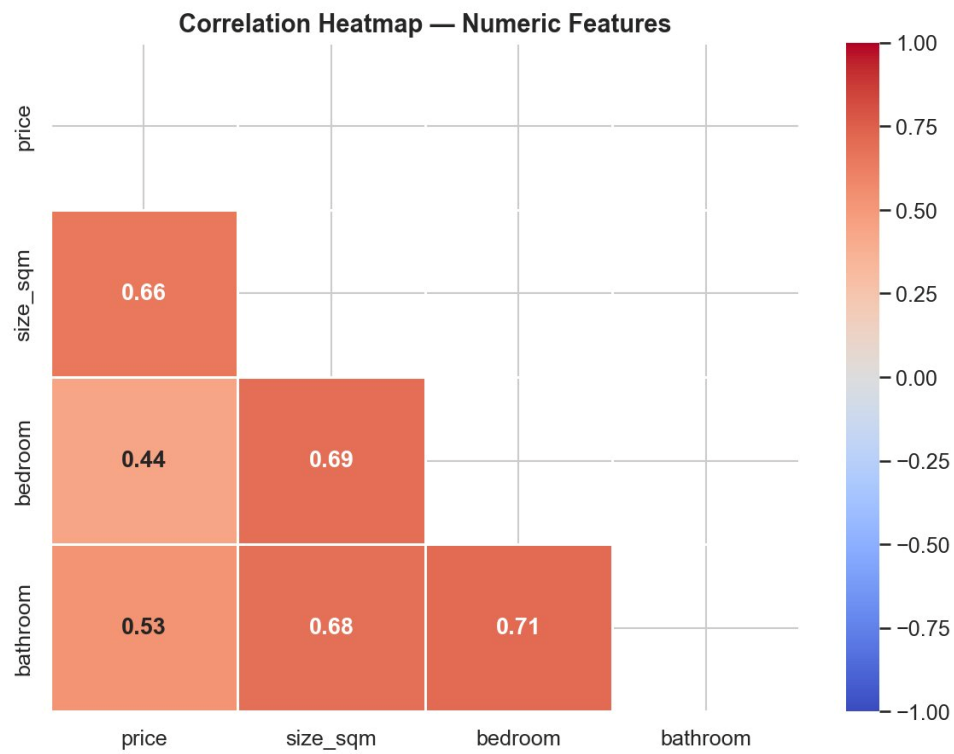


Figure 8: Pearson correlation heatmap — lower triangle shows pairwise correlations

5. Hypothesis Testing

5.1 Hypothesis 1 — Property Type and Price (T-Test)

Hypotheses

H_0 : There is no significant difference in mean price between Villas and Apartments in Cairo.

H_1 : Villas are significantly more expensive than Apartments in Cairo.

Statistical Test

An Independent Samples T-Test (T-Test, equal_var=False) was applied to compare the mean prices of 5,493 Apartment listings and 1,243 Villa listings. T-Test variant was chosen because the two groups have unequal sample sizes and different variances (Apartment std = 1.67M, Villa std = 4.19M).

Visualization

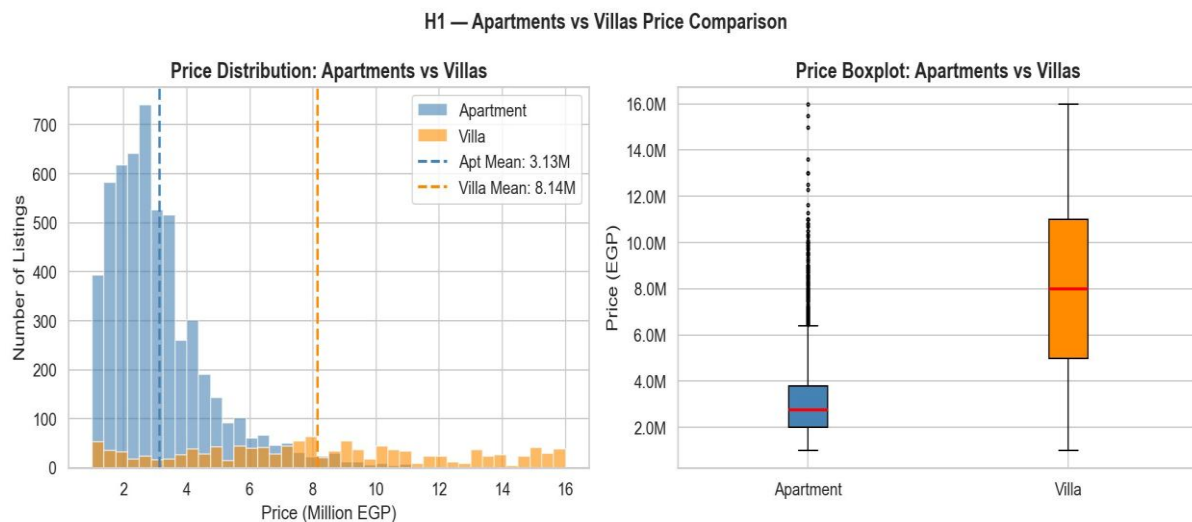


Figure 9: H1 — Overlapping price distributions (left) and boxplot comparison (right) for Apartments vs. Villas

Figure 9 clearly illustrates the separation between the two groups. The apartment price distribution is tightly clustered around 2–4M EGP, while the Villa distribution is wider and centers around 8M EGP. The two distributions barely overlap, providing strong visual evidence of a significant difference before the test is even run.

Results

Group	N	Mean Price	Std Dev
Apartment	5,493	3.13M EGP	1.67M EGP
Villa	1,243	8.14M EGP	4.19M EGP

T-Statistic	P-Value	Decision ($\alpha = 0.05$)
41.4857	< 0.000001	REJECT H_0

Interpretation

The T-statistic of 41.49 is exceptionally large, indicating the price difference between the two groups is 41 standard errors apart — far beyond the threshold for statistical significance. The p-value is effectively zero ($p < 0.000001$), meaning there is less than a 0.0001% probability that this difference occurred by chance. The mean price difference of 5.01M EGP between Villas and Apartments confirms that property type is a highly significant driver of residential property prices in Cairo. H_0 is rejected with overwhelming confidence.

5.2 Hypothesis 2 — Property Size and Price (Pearson Correlation)

Hypotheses

H_0 : There is no significant linear correlation between property size (m^2) and price.

H_1 : There is a significant positive linear correlation between property size and price.

Statistical Test

The Pearson Correlation Coefficient was calculated between size_sqm and price across all 9,209 clean listings. Pearson's r measures the strength and direction of the linear relationship between two continuous numeric variables, ranging from -1 (perfect negative) to +1 (perfect positive).

Visualization



Figure 10: H2 — Scatter plot with regression line (left) and average price by size range (right)

Figure 10 (left) shows the scatter plot of all 9,209 properties. The red regression line has a clear positive slope, visually confirming the upward relationship. The spread of dots widens with increasing size, indicating some heteroscedasticity — which is normal for real estate data where larger properties include both modest and luxury units. Figure 10 (right) groups properties into size ranges and shows a perfectly monotonic increase in average price from 2.03M EGP (under 100 m^2) to 8.84M EGP (350+ m^2).

Results

Pearson r	P-Value	Decision ($\alpha = 0.05$)
0.6558 (Moderate Positive)	< 0.000001	REJECT H_0

Interpretation

The Pearson correlation coefficient of $r = 0.66$ indicates a moderate-to-strong positive linear relationship between property size and price. This means that approximately 43% of the variance in price can be explained by size alone ($r^2 = 0.43$). The p-value is effectively zero, confirming the correlation is statistically significant and not due to chance. Properties under 100 m² average 2.03M EGP, rising to 8.84M EGP for properties over 350 m² — a 4.4x price increase. H_0 is rejected; property size is a statistically significant predictor of price.

5.3 Hypothesis 3 — Bedroom Count and Price (ANOVA)

Hypotheses

H_0 : There is no significant difference in mean price across properties with different bedroom counts.

H_1 : The number of bedrooms significantly affects property price.

Statistical Test

A One-Way ANOVA (Analysis of Variance) was applied to compare mean prices across 7 bedroom groups (0 through 6 bedrooms). Groups with 7 and 8 bedrooms were excluded due to insufficient sample sizes (fewer than 30 listings), which could compromise the reliability of the test. ANOVA tests whether the variance between groups is significantly greater than the variance within groups.

Group Summary

Bedrooms	N	Mean Price	Typical Property
0 (Studio)	35	1.97M EGP	Studio unit
1	133	2.29M EGP	Small apartment
2	1,391	2.71M EGP	Compact family apartment
3	5,210	4.05M EGP	Standard family apartment
4	1,891	6.72M EGP	Large apartment or townhouse
5	461	7.53M EGP	Villa or twin house
6	72	7.20M EGP	Large villa

Visualization

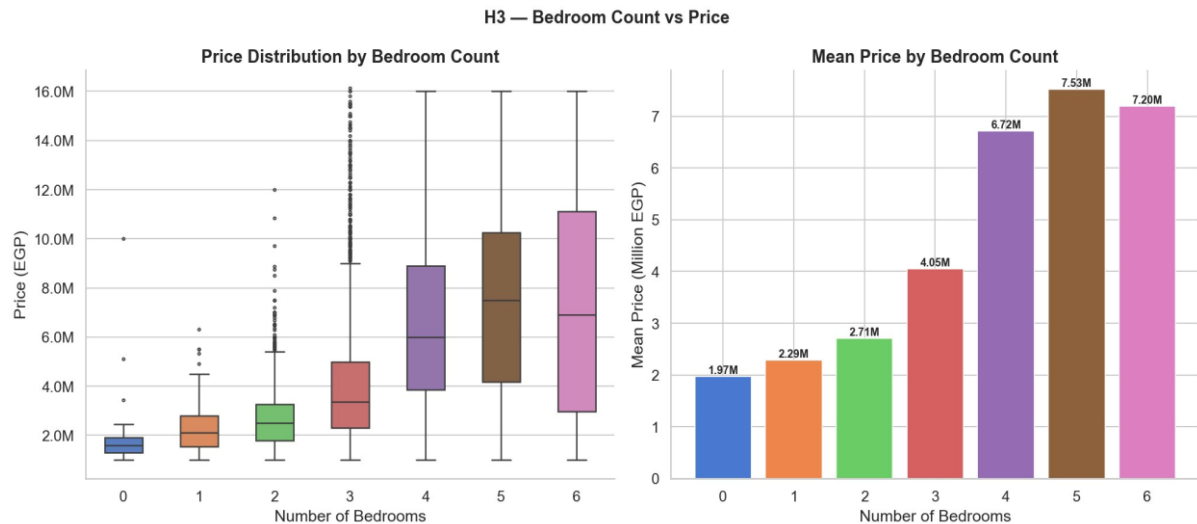


Figure 11: H3 — Price distribution by bedroom count (boxplot left, mean bar chart right)

Figure 11 clearly illustrates the price escalation with bedroom count. The boxplot (left) shows progressively higher and more spread-out price ranges as bedroom count increases. The bar chart (right) confirms the monotonic upward trend from Studio (1.97M) to 5-bedroom (7.53M EGP).

Results

F-Statistic	P-Value	Decision ($\alpha = 0.05$)
437.4324	< 0.000001	REJECT H_0

Interpretation

The F-statistic of 437.43 is extremely large, indicating that the variance in price between bedroom groups is 437 times greater than the variance within any individual group. The p-value is effectively zero. Mean prices increase from 1.97M EGP for Studios to 7.53M EGP for 5-bedroom properties. H_0 is rejected; bedroom count has a statistically significant and practically meaningful effect on property price in Cairo.

6. Results Summary

The following table summarizes the results of all three hypothesis tests conducted in this study:

Hypothesis	Test	Statistic	Decision
H1: Villas vs Apartments	T-Test	t = 41.49	Reject H ₀
H2: Size vs Price	Pearson Corr	r = 0.66	Reject H ₀
H3: Bedroom Count vs Price	One-Way ANOVA	F = 437.43	Reject H ₀

All three null hypotheses were rejected at $\alpha = 0.05$ with p-values effectively equal to zero in all cases.

7. Conclusions

This study analyzed 9,209 residential property listings from Cairo, Egypt, to identify the key factors influencing property prices. Three hypotheses were tested using formal statistical methods. The following conclusions are drawn:

7.1 Property Type is a Major Price Driver

Property type has a highly significant effect on price in Cairo's residential market. Villas command an average price of 8.14M EGP compared to 3.13M EGP for Apartments — a difference of 5.01M EGP. The T-test confirmed this difference is statistically significant with overwhelming confidence ($t = 41.49$, $p < 0.001$). Apartments account for 59.6% of the market, making them the dominant and most accessible property type, while Villas and Twin Houses represent the premium segment.

7.2 Property Size is a Strong Predictor of Price

A significant moderate-to-strong positive correlation was found between property size and price ($r = 0.66$, $p < 0.001$). Size alone explains approximately 43% of the variance in price. Properties smaller than 100 m² average 2.03M EGP, rising to 8.84M EGP for properties above 350 m² — a 4.4x increase. This confirms that size is the most important continuous numeric predictor of residential property price in Cairo.

7.3 Bedroom Count Significantly Affects Price

The number of bedrooms significantly affects property price ($F = 437.43$, $p < 0.001$). Mean prices increase consistently from 1.97M EGP for Studios to 7.53M EGP for 5-bedroom properties. The strong F-statistic indicates the between-group variance is 437 times larger than the within-group variance, confirming that bedroom count is a meaningful and statistically significant determinant of price.

7.4 Overall Conclusion

In Cairo's residential property market, property type, size (m²), and bedroom count are all statistically significant drivers of price. Among these, property type produces the largest absolute price gap (Villas vs Apartments: 5.01M EGP difference), while size demonstrates the strongest continuous linear relationship with price ($r = 0.66$). These findings are consistent with real estate market theory and provide quantitative evidence for pricing dynamics in Cairo's housing market.

The results of this study can be valuable for homebuyers estimating fair market value, real estate investors identifying underpriced segments, and policymakers understanding affordability gaps in Cairo's housing market.

8. References & Tools

- Dataset: Egypt Cairo Housing Prices — iyadelwy (Kaggle, 2024).
kaggle.com/datasets/iyadelwy/egypt-housing-prices
- Python 3 — Primary programming language
- Pandas & NumPy — Data manipulation and numerical analysis
- Matplotlib & Seaborn — Data visualization
- SciPy (stats module) — Statistical hypothesis testing (ttest_ind, pearsonr, f_oneway)
- Jupyter Notebook — Interactive analysis environment